

JHU Engineering for Professionals
Applied and Computational Mathematics
Data Mining: 625.740

Homework: Parametric Models for Unsupervised Learning

Using the dataset `unsupervised_learning_data.txt`:

Run k -means clustering, fuzzy k -means clustering, and expectation-maximization on these data. For each algorithm, use $k = 3$. You may also want to experiment with other values of k .

For each of the algorithms, report the resulting estimates of the means and covariances (where applicable).

Plot the resulting membership sets, plot the estimated means, and plot the estimated covariances (where applicable) for each run and briefly discuss the results.

Please make sure the coordinate axes are scaled equally. Where points have fuzzy set membership, be sure to show this by appropriate use of color.